

Detail Syllabus

B.Sc. (Data Science)

Semester-IV

Savitribai Phule Pune University
B.Sc. Data Science (2024 Pattern)
Sem-IV

DS-251-MJ-T: Relational Database Management System

No. of Credits: 2	Teaching Scheme Theory: 2 Hrs/Week	Examination Scheme Continuous Evaluation: 15 Marks End Semester : 35 Marks	
Prerequisites Basic Knowledge of DBMS, Knowledge of SQL Queries, Basics of relational design			
Objectives - To teach the essential concepts of RDBMS with PL/PgSQL - Understand the need of transaction processing and learn techniques for controlling the consequences of concurrent data access. - Learn how different recovery techniques restore the database to a consistent state after failures			
Course Outcomes On Completion of this course, student will be able to - CO1: Understand the basic syntax and structure of PL/SQL language. CO2: Analyze database transactions and can control them by applying ACID properties. CO3: Demonstrate the basic concepts of transaction processing and concurrency control. CO4: Understand concurrent transactions affect the recovery process. CO5: Analyze various databases.			
Unit No.	Name of Unit	Teaching Hours	CO Targeted
1	Introduction to PL/SQL	10	CO1
PL/SQL: Structure, Elements, Data types, Control Structure, Iterative Control, Views, Cursors - Stored Procedure and Function, Exceptional Handling and Errors, Triggers.			
2	Transaction Management	5	CO2
Define Transaction, Properties of Transaction, States of Transaction , The problems associated with concurrently executing transactions What is Schedule Types of Schedule Concept of Serializability, Precedence graph for serializability			
3	Concurrency Control Techniques	7	CO3
Ensuring serializability by locks, Lock Based Protocols ,Different locks modes, 2PL & it's Variation, Timestamp Based Protocol, Thomas Writes rule, Multiple Granularity Phantom problem, Deadlock & Deadlock handling technique.			

4.	Database Recovery Technique	6	CO4
Recovery concepts, Log based Recovery technique (Deferred and Immediate), Checkpoints, Shadow paging, Aries Algorithm, Recovery with concurrent Transactions. Database backup and Recovery from Catastrophics failures.			
5	Advanced Topics in Database	2	CO5
Parallel Database, Distributed Database, Multimedia Database , Mobile Database , Web Database , Multidimensional Database, Data Warehouse , OLTP Vs OLAP, Bigdata, NoSQL Databases			
Reference Books			
1. Book1 - Database Management Systems, Raghu Ramakrishnan and Johannes Gehrke 2. Book2 - “Fundamentals of Relational Database Management System”, Sumathi, S. Esakkirajan 3. Book 3 - Database System Concepts by Henry Korth and A. Silberschatz 4. Book 4 - Practical Postgresql , By Joshua D. Drake, John C Worsley (O’Reillypublications) 5. Book 5 - PostgreSQL, Korry Douglas, ISBN:9780672327568, Sams			

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DS-252-MJ-T : Data Structure II

No. of Credits: 2	Teaching Scheme Theory: 2 Hrs./Week	Examination Scheme Continuous Evaluation: 15 Marks End Semester : 35 Marks	
Prerequisites • Basic knowledge of Python Programming. • Fundamental Knowledge of Array, Linked List, Stack, Queue Data Structure • Familiarity with basic Searching and Sorting techniques			
Objectives • To understand advanced data structures and their applications in Data Science. • To analyze and implement complex algorithms using Python. • To improve problem-solving skills by applying data structures to real-world problems. • To introduce students to efficient searching, sorting, and graph-based techniques. • To enhance the ability to optimize performance in large-scale data-driven applications			
Course Outcomes On Completion of this course, student will be able to - CO1: Understand the Fundamental Concepts of Non-Linear Data Structures CO2: Develop optimized algorithms for data science applications CO3: Apply graph algorithms for solving real-world problems. CO4: Evaluate the efficiency of algorithms using Time and Space complexity.			
Unit No.	Name of Unit	Teaching Hours	CO Targeted
1	Tree	10	CO1
Concept and Terminologies Types of Binary Tree-Binary Tree, Skewed tree, Strict Binary Tree, Full Binary Tree, Complete Binary Tree, Expression tree, Binary Search Tree, Heap			
Representation Static and Dynamic Representation of tree			
Implementation and Operations on Binary Search Tree Create, Insert, Delete, Search Tree Traversal-preorder, inorder, postorder ,Level order traversal using queue Counting Leaf node, Non leaf nodes, Copy, Mirror			
Application of Trees Heap Sort Introduction to Greedy Strategy, Huffman encoding (Implementation Using Priority Queue)			
Applications of tree in Data Science Decision tree in machine learning			

Hierarchical clustering in data analysis			
2	Efficient Search Tree	4	CO1,CO2
Terminology- Balanced trees-AVL Trees, Red Black tree, Splay Tree, Lexical Search tree-Trie, Multiway search Trees-B and B+ tree AVL Tree- Concept and Rotations			
3	Graph	10	CO1,CO2,CO3, CO4
Concept and Terminology Graph representation-Adjacency Matrix, Adjacency List Graph Traversals-Breadth First Search, Depth First Search (with implementation) Applications of Graph - Topological Sorting Use of Greedy Strategy in Minimal Spanning Trees(Prim's and Kruskal Algorithm) Single Source Shortest Path-Dijkstra's Algorithm Dynamic Programming Strategy, All Pair shortest Path(Floyd Warshall Algorithm) Applications of Graph in Data Science Use of Graph in Social Network Graph based Machine Learning Transportation & Route Optimization			
4	Hash table	6	CO1, CO2
Concept of Hashing Terminologies-Hash Table, Hash Function, Bucket, Hash address, Collision, Synonym, Overflow Properties of good Hash function Hash Function-Division function, MID Square, Folding Method Collision Resolution Techniques Open Addressing-Linear Probing, Quadratic Probing, Rehashing Chaining: Coalesced, Separate Chaining Applications of Hash Table in Data Science Efficient Lookup in Datasets : Searching specific values or attributes in large datasets Counting Frequencies : Counting frequencies of words in large datasets Caching Results : Storing computed results of expensive operations for reuse			
Reference Books			
5. Data Structures and Algorithms in Python Authored by Michael T. Goodrich, Roberto Tamassia, and Michael H. Goldwasser 6. Data Structures Using Python by Shriram K. Vasudevan, 7. Data Structures and Algorithm Analysis in Python by Mark Allen Weiss 8. Python Data Structures and Algorithms by Benjamin Baka			

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DS-253-MJ-P : Lab Course on DS-251-MJ-T and DS-252-MJ-T
(Relational Database Management System and Data Structure-II)

No. of Credits: 2	Teaching Scheme Practical: 4 Hrs./Week	Examination Scheme Continuous Evaluation: 15 Marks End Semester : 35 Marks
Course Objectives - To solve real world computational problems. - To perform operations on relational database management systems. - Understand the Fundamental Concepts of Non-Linear Data Structures - Develop optimized algorithms for data science applications		
Course Outcomes On completion of this course students will able to: CO1: To use PostgreSQL (PL/SQL.) CO2: To perform advanced database operations (View ,Function, cursor, Trigger) CO3: Implement Fundamental Concepts of Non-Linear Data Structures CO4: Implement nonlinear Data Structure dynamically with their application		
Operating Environment: For DBMS: - Operating System: Linux or any relevant Operating System - DBMS: PostgreSQL For Data Structures: - Python is cross-platform and can run on: Windows (Windows 10/11) Linux (Ubuntu, Fedora, Debian, etc.) macOS (Latest versions) To write and execute Python programs efficiently, you can use: VS Code (Lightweight, supports extensions) PyCharm (Best for large projects, auto-suggestions) Jupyter Notebook (Great for data visualization & step-by-step execution) IDLE (Comes pre-installed with Python, good for beginners) Spyder (Best for scientific computing)		
Course Contents:		
Relational Database Management Systems Assignments		
Assignment 1: Views		

1) To Create and Drop Views
Assignment 2 : Stored Procedure
1) Simple Stored Procedure 2) Stored Procedure with IN, OUT and IN/OUT parameter
Assignment 3: Stored Function
1) Simple Stored Function 2) Stored Function that returns
Assignment 4 : Cursors
1) Simple Cursor 2) Parameterize Cursor
Assignment 5 : Exception Handling
1) Simple Exception- Raise Debug Level Messages 2) Simple Exception- Raise Notice Level Messages 3) Simple Exception- Raise Exception Level Messages
Assignment 6 : Triggers
1) Before Triggers (insert, update, delete) 2) After Triggers (insert, update, delete)
Assignments for Data Structures – II
Assignment 1: Binary Search Tree and Traversals
1. Implement Binary Search Tree (BST) to perform following operations on BST– Create, Recursive Traversals - Inorder, Preorder, Postorder 2. Perform following operations: insert, delete
Assignment 2: Binary Search Tree Operations
1. Implement Binary Search Tree (BST) to perform following operations on BST– Copy and mirror image of BST, counting leaf, non-leaf and total nodes. 2. Level-order traversal of binary search tree using queue.
Assignment 3: Applications of Binary Tree
1. Sort set of elements using Heap sort 2. Encode a set of characters using Huffman encoding
Assignment 4: Graph implementation
1. Implement Graph as adjacency matrix and adjacency list 2. Calculate indegree and out degree of vertices 3. Graph traversals: BFS and DFS.
Assignment 5: Graph Applications - I
1. Implementation of Topological sorting 2. Implementation of Prims/Kruskals Minimum spanning tree algorithm
Assignment 6: Graph Applications - II
1. Implementation of Dijkstra's shortest path algorithm for finding Shortest Path from a given source vertex using adjacency cost matrix. 2. Implementation of Floyd Warshall algorithm for all pairs shortest path.
Assignment 7: Hash Table
1. Implementation of static hash table with Linear Probing. 2. Implementation of static hash table with chaining

Books
1. Database Management Systems, Raghu Ramakrishnan and Johannes Gehrke 2. “Fundamentals of Relational Database Management System”, Sumathi, S. Esakkirajan 3. Database System Concepts by Henry Korth and A. Silberschatz 4. Practical Postgresql , By Joshua D. Drake, John C Worsley (O'Reillypublications) 5. PostgreSQL, Korrry Douglas, ISBN:9780672327568, Sams 6. Data Structures and Algorithms in Python Authored by Michael T. Goodrich, Roberto Tamassia, and Michael H. Goldwasser, 7. Data Structures Using Python by Shriram K. Vasudevan, 8. Data Structures and Algorithm Analysis in Python by Mark Allen Weiss 9. Python Data Structures and Algorithms by Benjamin Baka

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DS-271-VSC-P: Data Analytics

No. of Credits: 02	Teaching Scheme Practical: 4 Hours/Week	Examination Scheme Continuous Evaluation: 15 Marks End Semester: 35 Marks
Prerequisites • Basic knowledge of logic and Python programming concepts • Knowledge of problem-solving tools like algorithms, flowcharts and pseudo codes will be an added advantage		
Objectives • To explore the fundamental concepts of data analytics. • To understand the various methods and tools for data analytics. • Design a simple database. • Query a database using SQL in Python		
Course Outcomes On Completion of this course, student will be able to - CO1: Use different data analytics functions to solve a problem. CO2: Apply different data cleaning functions on a given data. CO3: Explain basic excel formulas and functions and apply them on a data set. CO4: Design a simple database with DDL and DML commands. CO5: Write sub queries and join operations for retrieving data from various tables. CO6: Use the aggregations and group operations for data analysis in python.		
Guidelines: Operating Environment: Excel, PostgreSQL, Python		
List of Assignments:		
Excel: Assignment 1 : Getting Started with Excel • Creation of spread sheets, Insertion of rows and columns, Drag & Fill, use of Aggregate functions. Assignment 2 : Working with Data • Importing data, Data Entry & Manipulation, Sorting & Filtering, Data Validation, Pivot Tables & Pivot Charts Assignment 3 : Data Analysis Process • Conditional Formatting, What-If Analysis, Data Tables, Charts & graphs Assignment 4 : Cleaning Data with Text Functions and Cleaning Data Containing Date and Time Values • Use of UPPER and LOWER, TRIM function, Concatenate, use of DATEVALUE function, DATEADD and DATEDIF, TIMEVALUE functions. Assignment 5 : Conditional Formatting • Formatting, parsing, and highlighting data in spreadsheets during data analysis.		

Assignment 6 : Working with Multiple Sheets

- Work with multiple sheets within a workbook is crucial for organizing and managing data, perform complex calculations and create comprehensive reports

Assignment 7 : Generation of report & presentation using Autofilter & macro**SQL (Using Python)**

Connect to PostgreSQL database using Python and Write SQL queries.

- Filtering and sorting data (SELECT, WHERE, ORDER BY), Aggregation (SUM(), AVG(), COUNT()), Joining multiple tables (JOIN), Fetch results and analyze them in Pandas.

Python:**Assignment 1: Load a dataset and perform some primary operations.**

- info(), describe(), shape, size, loc, sort_values, value_counts() used to describe the dataframe, Handle missing values, Remove duplicate records, Standardize column names
- Getting introduced to essential packages like NumPy, SciPy, and pandas from the Data Analytics point of view.

Assignment 2 : Basic statistical operations

- Apply basic statistical operations on a dataset- Compute the mean, median, mode, range, quartiles, and variance for one or more attributes.

Assignment 3 : Data pre-processing

Apply data pre-processing techniques

- Describing data set
- Shape the data set
- Displaying specific rows from the data set
- Partition them into appropriate number of bins by equal-frequency as well as equal-width partitioning.

Assignment 4 : Data Visualization

- View the data using various 2-D, 3-D plots and charts to detect and handle outliers

Reference Books

1. How to solve it by Computer, R.G. Dromey, Pearson Education.
2. Programming Python, O'Reilly, 4th Edition, 2010

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DS-281-FP : Mini Project
(Software Engineering Project)

No. of Credits: 2	Teaching Scheme Practical: 4 Hrs/Week	Examination Scheme Continuous Evaluation: 15 Marks End Semester : 35 Marks
Prerequisites Fundamental Knowledge of Software Engineering, RDBMS		
Objectives - Apply software engineering principles to design, develop a small-scale software project. - Demonstrate understanding of software development life cycles, methodologies, and tools. - Develop teamwork, communication and problem-solving skills.		
Course Outcomes On Completion of this course, student will be able to - CO1: Understand software engineering discipline. CO2: The project should have a clear set of requirements, design documents. CO3: The project can be developed using any process model.		
Unit No.	Name of Unit	
1	Project Proposal	
	- Project title and description - Project goals and objectives - Technical specifications and requirements	
2	Problem Definition and Scope	
	- Problem Description - Study of Existing system(Manual or computerized) - Drawbacks of Existing system - Scope of the Proposed System	
3	Feasibility Study	
	- Technical Feasibility - Economical Feasibility - Operational Feasibility	
4	Fact Finding Techniques	
	- Interview - Questionary - Survey - Observations	

5	Entity Relationship Diagram
	• Identify entities & attributes • Identify relations in entities
6	UML Diagrams
	• Class Diagram • Use case Diagram • Sequence Diagram • Activity Diagram • Component Diagram • Deployment Diagram
7	Data Dictionary
	• Designing the Database • Normalized to 3NF
8	Designing queries related to Functional requirements
	• Simple Queries • Nested Queries(using Aggregate Functions) • Views • Functions • Cursors • Trigger
Reference Books	
1. Book1:Software Engineering”, by Roger Pressman, 8 th Edition 2. Book2: Practical PostgreSQL”, O’Reilly Publications	

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DS-291-MN-T : Testing of Hypothesis and Sampling Distributions

No. of Credits: 02	Teaching Scheme Theory: 2 Hours/Week	Examination Scheme Continuous Evaluation:15 Marks End Semester : 35 Marks	
Prerequisites Basics of Distribution			
Objectives - To study the hypothesis - To perform the large sample test - To overview the sampling distribution - To understand the difference between small sample and large sample test - To study hypothesis testing based on sampling distributions			
Course Outcomes On Completion of this course, student will be able to – CO1: Identify the difference between statistic, estimate and parameter. CO2: Formulate the null and alternative hypotheses CO3: Calculate type-I and type-II error, p -value CO4: Apply small, large sample tests in real life problems CO5: Derive probability distribution function of chi-square, t, F distribution CO6: Explains interrelation between the above distributions and their properties. CO7: Get familiar with statistical tests of hypothesis and are able to apply in real life situations in various fields			
Unit No.	Name of Unit	Teaching Hours	CO Targeted
1	Testing of Hypothesis	08	CO1, CO2, CO3
Statistic, estimate and parameter. Sampling distribution of a statistic, standard error of a statistic with illustrations. Statistical Inference: Introduction to problem of Estimation and testing of hypothesis. Difference between estimator and estimate. Concept of unbiased estimator. Point and interval estimation. Statistical hypothesis, null and alternative hypothesis, simple and composite hypothesis, one sided and two-sided alternative hypothesis, critical region, type-I and type-II error, level of significance, p -value. Two-sided confidence interval. Tests of hypotheses using i) critical region approach, ii) p -value approach and iii) confidence interval approach. Tests for population means (large sample / approximate tests): Testing of population mean (μ) with a specified value (μ_0) for the known variance. Testing equality of two population means ($\mu_1 = \mu_2$) for the known variance. Construction of two-sided confidence interval for population mean μ and for the mean difference $\mu_1 - \mu_2$.			

Testing of population proportion (P) with a specified value (P_0). Testing equality of two population proportions ($P_1 = P_2$). Construction of two-sided confidence interval for P and $P_1 - P_2$.			
2	Chi-square Distribution	08	CO5
Definition of chi-square random variable as a sum of squares of independent and identical standard normal variables. Derivation of the p.d.f. of Chi-square variable with n degrees of freedom (d.f.) using MGF. Mean, variance, MGF, CGF, central moments skewness, kurtosis, mode, additive property. Use of chi-square tables for calculations of probabilities. Normal approximation: $\frac{\chi_n^2 - n}{\sqrt{2n}}$ (statement only) Distribution of $\underline{X}$ and $\frac{nS^2}{\sigma^2} = \frac{1}{\sigma^2} \sum_{i=1}^n (X_i - \underline{X})^2$ for a random sample from a normal distribution using orthogonal transformation, independence of $\underline{X}$ and S^2 .			
3	t-Distribution	04	CO5
Definition of r.v. with n d.f. in the form of $= \frac{U}{\sqrt{\frac{V}{n}}}$, where $U \sim N(0,1)$ and V is chi-square with n d.f., where U and V are independent random variables. Derivation of the p.d.f of distribution, nature of probability curve, mean, variance, moments, mode. Use of t-tables for calculations of probabilities, Normal approximation (statement only).			
4	Snedecore's F-Distribution	05	CO5
Definition of r.v. with n_1 and n_2 d.f. as $F_{n_1, n_2} = \frac{X_1 n_1}{X_2 n_2}$ where X_1 and X_2 are independent chi-square variables with n_1 and n_2 d.f. Derivation of the p.d.f, nature of probability curve, mean, variance, moments, mode. Distribution of $\frac{1}{F_{n_1, n_2}}$, use of tables for calculation of probabilities. Interrelationship between Chi-square, t and F -distributions.			
5	Tests based on Sampling Distributions	05	CO6, CO7
Tests based on chi-square distribution: Test for independence of two attributes, Test for goodness of fit, Test for variance for known and unknown mean. Tests based on t-distribution: Tests for population means, Paired t-test for one-sided and two-sided alternatives. Test based on F-distribution: Test for $H_0: \sigma_1^2 = \sigma_2^2$ against one-sided and two-sided alternatives when i) means are known and ii) means are unknown. Take $F = \frac{S_1^2}{S_2^2}$.			
Reference Books			
1. Goon, A. M., Gupta, M. K. and Dasgupta, B. (1983). Fundamentals of Statistics, Vol. 1, Sixth Revised Edition, The World Press Pvt. Ltd., Calcutta. 2. Gupta, S. C. and Kapoor, V. K. (2002), Fundamentals of Mathematical Statistics, (Eleventh Edition), Sultan Chand and Sons, 23, Daryaganj, New Delhi, 110002. 3. Kulkarni, M. B., Ghatpande, S. B. and Gore, S. D. (1999), Common Statistical Tests, Satyajeet Prakashan, Pune 411029 4. Medhi, J., Statistical Methods, Wiley Eastern Ltd., 4835/24, Ansari Road, Daryaganj, New Delhi – 110002. 5. Meyer, P. L., Introductory Probability and Statistical Applications, Oxford and IBH Publishing Co. New Delhi. 6. Mood, A. M. and Graybill, F. A. and Boes D.C. (1974). Introduction to the Theory of Statistics, Ed. 3, McGraw Hill Book Company.			

7. Mood, A. M., Graybill F. A. and Bose, F. A. (1974), Introduction to Theory of Statistics (Third Edition, Chapters II, IV, V, VI), McGraw - Hill Series G A 276
8. Neil, A. Weiss, (2016). Introductory Statistics, Tenth Edition, Pearson.
9. Purohit, S. G., Gore S. D., Deshmukh S. R. (2008). Statistics Using R, Narosa Publishing House, New Delhi.
10. W. and Cochran W. G. (1989). Statistical Methods, Eighth Ed. East West Press.

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DS-292-MN-P : Lab Course on DS-291-MN-T
(Testing of Hypothesis and Sampling Distributions)

No. of Credits: 2	Teaching Scheme Practical: 4 Hrs/Week	Examination Scheme Continuous Evaluation: 15 Marks End Semester : 35 Marks
Prerequisites Basics of Distribution		
Objectives Practical Implementation of - Hypothesis - Performing the large sample test - The sampling distribution - The difference between small sample and large sample test - Hypothesis testing based on sampling distributions		
Course Outcomes: At the end of this course, students will be able to: CO1: To draw conclusions and determining probability of making errors in hypothesis tests using critical values. CO2: To conduct various tests of significance like averages, population proportions, independence of attributes, variance etc.		
List of Assignments		
Sr. No.	Assignment	
1	Test for proportions (one sample problem and two sample problem)	
2	Test for means (large sample test).	
3	Test for means (one sample problem) (small sample test).	
4	Test for means (two sample problem) (small sample test).	
5	Paired t-test.	
6	Test for independence of attributes	
7	χ^2 goodness of fit test.	
8	Test based on χ^2 distribution : $H_0: \sigma^2 = \sigma_0^2$ for μ unknown	
9	Tests for population variance (two sample problem only).	
Reference Books		
19. Goon, A. M., Gupta, M. K. and Dasgupta, B. (1983). Fundamentals of Statistics, Vol. 1, Sixth Revised Edition, The World Press Pvt. Ltd., Calcutta.		

20. Gupta, S. C. and Kapoor, V. K. (2002), Fundamentals of Mathematical Statistics, (Eleventh Edition), Sultan Chand and Sons, 23, Daryaganj, New Delhi, 110002.
21. Kulkarni, M. B., Ghatpande, S. B. and Gore, S. D. (1999), Common Statistical Tests, Satyajeet Prakashan, Pune 411029
22. Medhi, J., Statistical Methods, Wiley Eastern Ltd., 4835/24, Ansari Road, Daryaganj, New Delhi – 110002.
23. Meyer, P. L., Introductory Probability and Statistical Applications, Oxford and IBH Publishing Co. New Delhi.
24. Mood, A. M. and Graybill, F. A. and Boes D.C. (1974). Introduction to the Theory of Statistics, Ed. 3, McGraw Hill Book Company.
25. Mood, A. M., Graybill F. A. and Bose, F. A. (1974), Introduction to Theory of Statistics (Third Edition, Chapters II, IV, V, VI), McGraw - Hill Series G A 276
26. Neil, A. Weiss, (2016). Introductory Statistics, Tenth Edition, Pearson.
27. Purohit, S. G., Gore S. D., Deshmukh S. R. (2008). Statistics Using R, Narosa Publishing House, New Delhi.
28. W. and Cochran W. G. (1989). Statistical Methods, Eighth Ed. East West Press.

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SEC-251-DS-T: Software Engineering

No. of Credits: 2	Teaching Scheme Theory: 2 Hrs/Week	Examination Scheme Continuous Evaluation: 15 Marks End Semester: 35 Marks	
Prerequisites Fundamentals of programming language, Databases, ER Modeling			
Objectives 1. To get knowledge and understanding of software engineering discipline. 2. To learn analysis and design principles for software project development. 3. To become familiar with software development life cycle models and their applications. 4. To understand software testing strategies and techniques.			
Course Outcomes On Completion of this course, student will be able to - CO1: Understand software engineering discipline. CO2: Compare and choose a process model for a software project development. CO3: Identify requirements analyze, prepare models and Agile Models. CO4: Apply design concepts and metrics in software development. CO5: Implement testing strategies to ensure software quality.			
Unit No.	Name of Unit	Teaching Hours	CO Targeted
1	Software Engineering Fundamentals	3	CO1
1.1 Definition of Software and the evolving role of software 1.3 Changing nature of software 1.4 Software myths. 1.5 Software Process 1.5.1 The Process Framework 1.5.2 Umbrella Activities 1.5.3 Process Adaptation			
2	Software Development Life Cycle (SDLC)	5	CO2
2.1 Generic Process Model 2.2 Prescriptive Process Models 2.2.1 The Waterfall Model 2.2.2 Incremental Process Models 2.2.3 Evolutionary Process Models 2.2.4 Concurrent Models 2.2.5 The Unified Process 2.6 Reverse Engineering			
3	Agile Development	6	CO3
3.1 Agility and Agile Process			

3.2 Agility Principles 3.3 The Politics of Agile Development 3.4 Human Factors 3.5 Extreme Programming (XP) 3.5.1 XP Values 3.5.2 XP Process 3.6 Adaptive Software Development (ASD) 3.7 Scrum 3.8 Dynamic System Development Model (DSDM) 3.9 Agile Unified Process (AUP)			
4	Software Requirement Analysis and Design Using UML	12	CO3,CO4
4.1 Requirement Elicitation 4.2 Software requirement specification (SRS) 4.3 Building the Analysis Model 4.4 Negotiating Requirements 4.5 Validating Requirements 4.6 Introduction to UML 4.7 Structural Modeling 4.7.1 Use case model 4.7.2 Class model 4.8 Behavioral Modeling 4.8.1 Sequence model 4.8.2 Activity model 4.8.3 Collaboration model 4.9 Architectural Modeling 4.9.1 Component model 4.9.2 Deployment model			
5	Software Testing	4	CO5
5.1 Introduction to Software Testing 5.2 Software Testing Life Cycle (STLC) 5.3 Software Testing Strategies & Techniques 5.3.1 Black box and White box Testing 5.4 Level of Testing 5.4.1 Unit Testing 5.4.2 Integration Testing 5.4.3 System Testing			
Reference Books			
1. Software Engineering : A Practitioner's Approach - Roger S. Pressman, McGraw hill(Eighth Edition) ISBN-13: 978-0-07-802212-8, ISBN-10: 0-07-802212-6 2. The Unified Modeling Language Reference Manual - James Rumbaugh, Ivar Jacobson, Grady Booch ISBN 0-201-30998-X 3. Effective Methods of Software Testing, William E Perry, 3rd Edition, Wiley Publishing Inc			

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Course Type: GE/OE. Course Code: OE-251-DS-T

Course Title: Ecommerce II

(To be offered to faculty other than Science and Technology)

Teaching Scheme 02 Hours /Week	No. of Credits 2	Examination Scheme Continuous Evaluation: 15 Marks End Semester: 35 Marks
Prerequisites • Knowledge of basic e-commerce concepts, business models and payment systems. • Understanding of digital marketing, financial systems, and internet technologies		
Course Objectives • To understand the technical and security aspects of e-commerce. • To explore data-driven decision-making and analytics in e-commerce. • To study supply chain and logistics management in e-commerce. • To gain insights into global e-commerce trends and challenges. • To learn about the integration of AI, Blockchain, and Cloud Computing in e-commerce.		
Course Outcomes On completion of the course, student will be able to— CO1: Implement secure e-commerce transactions and protect user data. CO2: Apply analytics tools to track and enhance e-commerce performance. CO3: Manage e-commerce logistics and understand global trends. CO4: Use emerging technologies such as AI, Blockchain, and Cloud for e-commerce applications.		
Course Contents		
Chapter 1	E-Commerce Logistics and Supply Chain Management	6 Hours
1.1. E-Commerce Supply Chain Overview 1.2. Inventory Management in E-Commerce 1.3. Role of Warehousing and Fulfillment Centers 1.4. Last-Mile Delivery Challenges 1.5. Reverse Logistics and Return Policies 1.6. Green and Sustainable Supply Chains		
Chapter 2	Data Analytics and Consumer Behavior in E-Commerce	7 Hours
2.1. Importance of Data Analytics in E-Commerce 2.2. Key Performance Indicators (KPIs) and Metrics		

2.3. Web Analytics (Google Analytics, Heatmaps, A/B Testing)		
2.4. Predictive Analytics & Customer Insights		
2.5. Recommendation Engines & Personalization		
2.6. Conversion Rate Optimization (CRO)		
2.7. Fraud Detection Using AI in E-Commerce		
Chapter 3	E-Commerce Security and Privacy	6 Hours
3.1. Importance of Security in E-Commerce		
3.2. Threats to E-Commerce (Phishing, Fraud, Cyber Attacks)		
3.3. Cryptography & Secure Transactions (SSL/TLS, Encryption)		
3.4. Digital Signatures & Certificates		
3.5. Firewalls & Intrusion Detection Systems		
Chapter 4	Advanced E-Commerce Technologies	5 Hours
4.1. Machine Learning for Product Recommendations		
4.2. Blockchain for Secure Transactions and Smart Contracts		
4.3. Cloud Computing and SaaS Platforms for E-Commerce		
4.4. The Role of 5G in E-Commerce Growth		
Chapter 5	Global Trends of E-Commerce	6 Hours
5.1. Cross-Border E-Commerce and Global Expansion		
5.2. Mobile Commerce (M-Commerce) Innovations		
5.3. Subscription-Based E-Commerce Models		
5.4. Social Commerce (Instagram, Facebook Shops, TikTok Commerce)		
5.5. Ethical and Sustainable E-Commerce Practices		
5.6. Future Challenges in E-Commerce		
Reference Books:		
1. E-Commerce Analytics: Analyze and Improve the Impact of Your Digital Strategy, Judah Phillips, Pearson		
2. Cloud Computing: Principles and Paradigms, Rajkumar Buyya, James Broberg, Andrzej Goscinski, Wiley		
3. Global E-Commerce: Theory and Case Studies, Jie Lin, Fei Gao, Springer		
4. Logistics and Supply Chain Management, Martin Christopher, Pearson		

Savitribai Phule Pune University
B.Sc. Data Science (2024 Pattern)
Sem-IV

Course Type: GE/OE Course Code: OE-252-DS-T
Course Title: Web Design II
(To be offered to faculty other than Science and Technology)

Teaching Scheme 02 Hrs/ week	No. of Credits 2	Examination Scheme Continuous Evaluation: 15 Marks End Semester: 35 Marks
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Prerequisites

- Knowledge of other programming languages will enhance your abilities with HTML and web development.
- Knowledge and understanding of Internet.

Course Objectives

- To learn to design Style sheets.
- To know to link and publish Web pages.
- To prepare the learners with the fundamentals of HTML programming and scripting languages.
- Learners should know how to create HTML documents, applying styles using CSS.

Course Outcomes

On completion of the course, student will be able to–

- Learn and use the CSS to design Web Pages.
- Link and publish Web pages.
- Develop and create a webpage by planning, designing, and implementing everything themselves.
- Understand and develop a dynamic web pages using JavaScript (client side programming) .

Course Contents

Chapter 1	CSS	8
1.1. Introduction of CSS and its Syntax 1.2. Ways to Insert CSS and Background image handling 1.3. Background colour management using CSS 1.4. Text and Font management using CSS 1.5. Managing Hyperlinks and List using CSS 1.6. Designing Borders and Outline 1.7. Setting Page Margin using CSS		
Chapter 2	XML	8
2.1. XML Namespaces and Infoset and Document Type 2.2. Definitions (DTDs)		

2.3.XML Schemas and XML-Parser		
2.4.Data Modeling , Document and Object Model (DOM)		
2.5.Displaying XML with XSLT		
Chapter 3	Introduction to JavaScript	8
3.1. Concept of Script, Types of Scripts		
3.2. Introduction to JavaScript.		
3.3. Variables, identifier and Operator, Control structure.		
3.4. Examples on JavaScript Operators.		
3.5. Functions		
3.6. Event Handling in JavaScript with examples.		
Case Study		6
Case study 1: Creation of forms, small case study to create HTML pages using all the above learnt techniques.		
Case study 2: Redesigning the Website of a Small Business.		
Case study 3: Create a Styled Web Page for a Coffee Shop.		
Reference Books:		
1. Learning Web Design: A beginner's Guide To HTML, CSS, Javascript, and Web Graphics - Jennifer Robbins		
2. HTML5: The Missing Manual - Matthew MacDonald		
3. HTML and JavaScript – Ivan Bayross		
4. Mastering HTML, CSS &Javascript Web Publishing		
Reference Link:		
1. JavaScript Guide - JavaScript MDN		
2. JavaScript reference - JavaScript MDN		
3. XML.com		

Savitribai Phule Pune University B.Sc. Data Science (2024 Pattern) Sem-IV Course Type: GE/OE Course Code: OE-253-DS-T Course Title: Digital Marketing- II <i>(To be offered to faculty other than Science and Technology)</i>		
Teaching Scheme 02 Hours /Week	No. of Credits 2	Examination Scheme Continuous Evaluation: 15 Marks End Semester: 35 Marks
Prerequisites - Digital marketing requires creativity and problem-solving abilities. - Experience with social media platforms (Facebook, Instagram, Twitter, LinkedIn, etc.) is beneficial, as digital marketing		
Course Objectives - To understand Digital Marketing as the most powerful marketing tool. - To Learn to create digital marketing artworks. - To use social media sites like Facebook, Instagram, Twitter, LinkedIn, and others to raise sales, engage customers, and establish your brand.		
Course Outcomes On completion of the course, student will be able to— CO1: Students will be able to communicate marketing strategies and results effectively to stakeholders. CO2: Student will be able to assess and enhance digital marketing campaigns' return on Investment. CO3: Students will gain practical experience with industry-standard digital marketing tools. CO4: Student will be adept at using a variety of social media channels to create and interact with communities, raise awareness of a brand.		
Course Contents		
Chapter 1	Online Consumer Behaviour Analysis	8
1.1 Consumer Behavior 1.2 Segmentation and targeting online customers 1.3 Psychological Responses 1.4 Social Trends		
Chapter 2	Social Media Marketing	8
2.1. Social Media Sites 2.2. Influence of Social Media Marketing 2.3. Power of Social Media 2.4. Monetization through Social Media		
Chapter 3	Future of Digital Marketing	8
3.1. Use of Artificial Intelligence (AI) in Digital Marketing. 3.2. Common use of household gadgets for online marketing.		

3.3. Digital Marketing strategies.	
Case Study	6
Case Study 1 : Experiential Learning : Creating a website. Case Study 2 : Online Consumer Behavior Analysis for an E-Commerce Fashion Brand Case Study 3	
Reference Books:	
1. Digital Marketing : Nitin Kamat, Chinmay Kamat (Himalaya Publishing House) 2. "Digital Marketing for Dummies" by Ryan Deiss and Russ Henneberry 3. "Influence: The Psychology of Persuasion" by Robert B. Cialdini 4. "Social Media Marketing Workbook: How to Use Social Media for Business" by Jason McDonald	
Reference Links:	
1. https://www.socialmediaexaminer.com/ 2. https://www.marketingprofs.com/	

Savitribai Phule Pune University
B.Sc. Data Science (2024 Pattern)
Sem-IV

Course Type: GE/OE Course Code: OE-254-DS-T

Course Title: AI for Everyone- II

(To be offered to faculty other than Science and Technology)

Teaching Scheme: 2hours / week	No. of Credits: 2	Examination Scheme Continuous Evaluation: 15 Marks End Semester: 35 Marks
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Course Objectives: -

1. Understand the basics of artificial intelligence and its subfields.
2. Explore real-world applications of AI across different industries.
3. Gain insights into the ethical, social, and economic implications of AI.
4. Develop an appreciation for the potential of AI to drive innovation and transformation.

Course Outcomes: -On completion of the course, student will be able to–

CO1: To understand different types of AI Models

CO2: To understand content optimization using AI.

CO3: To understand Animations and motions in AI

CO4: To Understand uses of AI tools.

Course Contents

Unit 1	Advanced AI Fundamentals	6 hours	CO1
1.1 Deep Dive into AI, Machine Learning & Deep Learning 1.2 Types of AI Models: Generative AI, NLP, Computer Vision, Reinforcement Learning 1.3 Latest AI Trends: AGI, Large Language Models (LLMs), and multimodal AI 1.4 Exploring AI Frameworks & APIs: Open AI, Hugging Face, Google AI			
Unit 2	AI for Advanced Text & Content Creation	8 hours	CO2
2.1 AI for Long-form Writing & Reports 2.2 Automating Research & Citation Management 2.3 AI for SEO & Content Optimization 2.4 Using AI for Professional Emails & Business Writing			
Unit 3	Advanced AI for Image & Video Processing	8 hours	CO3
3.1 AI Image Generation Beyond Basics 3.2 Deepfake Technology & Ethical Concerns 3.3 AI Video Editing & Creation			

3.4 AI Animation & Motion Capture			
Unit 4	AI Tools	8 hours	CO4
4.1 Chat GPT (Open AI) 4.2 Google Gemini (Bard AI) 4.3 Canva, Beautiful. AI, Gamma, Slides AI 4.4 Rytr, Grammarly			
References: 1. Artificial Intelligence: A Modern Approach – Stuart Russell & Peter Norvig. 2. Practical AI for Business Leaders – Anand S. Rao 3. AI-Powered Automation Handbook – Will Kelly 4. AI for Content Creators: How to Use AI Tools for Writing and Marketing – Rob Lennon 5. Human Compatible: Artificial Intelligence and the Problem of Control – Stuart Russell 6. https://www.grammarly.com/ai/ai-writing-tools/article-writer			